

# Aditya Karnam Gururaj Rao

Senior Software Engineer

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## PROFESSIONAL SUMMARY

Senior Software Engineer with 12+ years across AI/ML, distributed systems, and cloud infrastructure. Tech Lead at Zefr, architecting **ZAiN**, a production AI agent platform spanning MCP servers, tool execution, evals, permissioning, and observability, serving 500+ internal and 10,000+ external users. Delivered \$350K+ in direct business value, including Meta measurement systems with 83% performance gains and vector search infrastructure over 100M records. Published AI researcher (IEEE, Frontiers, arXiv) and open-source maintainer with contributions accepted by Anthropic. Previously founded and scaled a SaaS platform serving 100+ clients.

## EMPLOYMENT HISTORY

**ZEFR** Jan 2022 – Present  
*Los Angeles, California, United States, promoted from Software Engineer to Senior Software Engineer across four levels*

**Senior Software Engineer, Tech Lead of Customer Experience** (Dec 2025 – Present)

- ◆ Architected **ZAiN**, Zefr's production AI agent platform, spanning agents, MCP servers, tool execution, evals, agent gateway, permissioning, session compaction, and chat interface, including MCP apps for Claude Code, Codex, and Microsoft Copilot 365; serving 500+ internal and 10,000+ external users at 30K+ requests/month in pilot.
- ◆ Built agent observability infrastructure and dashboards tracking top agent users, tool-call volume, and production behavior from agent logs, enabling reliability evaluation of live AI workflows.
- ◆ Lead distributed system design spanning Kafka-based campaign processing, Airflow orchestration, search infrastructure, and API gateway architecture (Apigee/n8n), delivering customer-facing and internal platform systems across AWS and GCP.
- ◆ Authored 20+ RFCs and led cross-team design reviews across 4 teams, mentoring 5+ engineers and setting long-term technical direction, engineering standards, and platform reliability.

**Software Engineer to Software Engineer III** (Jan 2022 – Dec 2025)

- ◆ Delivered \$350K+ in measurable business value across measurement, search, and reporting systems, including two company hackathon wins shipped to production.
- ◆ Built Meta measurement systems (Python, Kafka, Redshift) achieving 83% performance gains.
- ◆ Designed and scaled vector search infrastructure spanning 100M+ records (Elasticsearch).
- ◆ Developed, optimized, and maintained the inventory availability endpoint for digital properties, processing data across multiple Kafka topics on AWS (ECS, Lambda, Firehose, S3).
- ◆ Built a Redshift metrics pipeline publishing custom metrics to AWS CloudWatch and Datadog.

**DIGITAL SYSTEMS ARCHITECT** Jun 2021 – Jan 2022  
*ePlata, Edwardsville, Illinois, United States*

- ◆ Built NFT platform on Cardano (Plutus Smart Contracts).
- ◆ Architected KYC Remittance API integrating Western Union APIs.
- ◆ Spearheaded IPC-V remittance platform generating \$3M transaction volume.
- ◆ Delivered full-stack engineering for 8+ white-label fintech clients (React, TypeScript, GraphQL, Docker).

**FULL STACK ENGINEER** Feb 2020 – May 2021  
*ePlata, Edwardsville, Illinois, United States*

- ◆ Migrated AWS EC2 workloads to ECS to optimize resource utilization and reduce costs.
- ◆ Mentored interns and built GraphQL API testing templates.
- ◆ Rebuilt logging service with support for Sentry, Logzio + PII sanitization.

**SOFTWARE ENGINEER** Jun 2018 – Jan 2020  
*ePlata, United States*

- ◆ Integrated Splunk ML Toolkit for Payment Fraud System.
- ◆ Improved build pipelines with AWS CodeBuild (75% time reduction).
- ◆ Built native contacts integration (Swift, Kotlin, TypeScript) increasing load performance by 60%.
- ◆ Created CI/CD pipelines with Docker, Jenkins, CircleCI, AWS CodeDeploy.

**GRADUATE RESEARCH ASSISTANT** Aug 2017 – Jun 2018  
*University of Missouri–St. Louis, St. Louis, Missouri, United States*

- ◆ Thesis: *Efficient Reduced-Bias Genetic Algorithm for Generic Community Detection Objectives*, a novel genetic algorithm

- framework for clustering networks with 5M+ nodes.
- ◆ Collaborated with Oak Ridge National Lab, using NVIDIA Titan cluster computing resources on an Alzheimer's gene-expression network dataset.
- ◆ Reduced memory usage with a 3-dimensional bit-array container.

## FOUNDER & CTO

Aug 2014 – Aug 2016

*eventzé*

- ◆ Founded a SaaS platform for managing large college events covering registrations, merchandise, finance automation, and digital branding, partnering with 100+ college fests.
- ◆ Built and led a team of 10+ developers to scale the platform (PHP/CodeIgniter, ReactJS, REST APIs, payment gateway integration, DevOps automation).

## SELECTED PROJECTS & OPEN SOURCE

- ◆ **Call to Think**: personal essay blog on technology, AI, and society ([calltothink.com](https://calltothink.com)), built with Astro and deployed on Cloudflare Workers.
- ◆ **quecto**: a tiny OpenAI-compatible AI harness and coding agent in Rust with a ~1.3 MB core and ~3.5 MB agent binary.
- ◆ **embenx**: a universal embedding retrieval toolkit and agentic memory layer with a unified API for 15+ vector backends and MCP support for Claude and autonomous agents (`pip install embenx`).
- ◆ **locobench**: local coding model benchmarking, a Harbor and Ollama setup for running local agentic benchmarks with a leaderboard pipeline.
- ◆ **mcp-scholarly**: academic retrieval with MCP for scholarly search from AI assistants (185+ GitHub stars).
- ◆ **opencontext**: interoperable AI chat contexts across assistants ([open-context.dev](https://open-context.dev)).
- ◆ **medfit-llm**: applied model fine-tuning in healthcare.
- ◆ **google-drive-upload-git-action**: developer productivity automation as a reusable GitHub Action.

## EDUCATION

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|--------------------------------------------------------------------------|-------------|
| MS, <b>Global Business and Blockchain</b> , University of the Cumberland | 2021 – 2022 |
| MS, <b>Computer Science</b> , University of Missouri–St. Louis           | 2016 – 2018 |
| B.Tech, <b>Computer Science</b> , Vellore Institute of Technology        | 2012 – 2016 |

## PUBLICATIONS & RESEARCH

- ◆ *MEDFIT-LLM: Medical Enhancements through Domain-Focused Fine Tuning of Small Language Models*. IEEE, 2025.
- ◆ *CAG: Chunked Augmented Generation for Google Chrome's Built-in Gemini Nano*. arXiv:2412.18708, 2024.
- ◆ *AI-Enhanced Neural Network Framework for Cardiac Arrest Prognostics*. Frontiers in Health Informatics, 2024.
- ◆ *Automated Analysis of Glucose Response Patterns in Type 1 Diabetes Using Machine Learning and Computer Vision*. International Journal of Advanced Computer Science & Applications, 2025.
- ◆ *Neural Network-Enabled Diagnostic Framework for Correlating Ayurvedic Breathwork Protocols with Adaptive Skin Microbiome Dynamics*. Mathematical Modelling of Engineering Problems, 2025.
- ◆ *The Harness Engineer*. 2025.
- ◆ *Quantifying Adaptive Governance Success: A Mixed-Methods Analysis of 500 Organizations*. 2025.
- ◆ *Efficient Reduced-Bias Genetic Algorithm (ERBGA) for Generic Community Detection Objectives*. MS thesis, University of Missouri–St. Louis, 2018.
- ◆ *Transmission in Benes Network*. International Journal of Pure and Applied Mathematics, 2016.